Inuka Silva

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Highlighted Qualifications

- Level 2 Bachelor of Engineering, Co-op student, eligible for 4-month co-op positions starting in May 2027.
- Proficient in Python, Java, C/C++, and CAD software including Autodesk Inventor and Onshape.
- Experienced in collaborative engineering design cycles, from concept development and CAD modelling to 3D printing, fabrication, integration, and testing.

Education

Candidate for (B.Eng.), Mechatronics Engineering Co-op (3.63) Sept. 2025 – Present
McMaster University Hamilton, ON

- Transferred to Study Mechatronics/Robotics Engineering

Candidate for (B.S), Computer Science Sept. 2024 – April 2025
University of Toronto Mississauga, ON

Experience

Engineering and Computer Science Summer Intern May 2025 – Aug. 2025
REDPATH MINING North Bay, ON

- Designed and fabricated a data logger to monitor machine health and track machine lifecycle data.
- Integrated IMU sensors, LoRa, Arduino, and Raspberry Pi to enable wireless data collection and transmission.
- Created technical documentation outlining code functionality, assembly, and integration for future development.

Projects

ENGINEER 1P13 Client Solution | Autodesk Inventor Jan. 2025 – April 2025

- Designed and fabricated a 3D-printed watering can, leveraging air pressure to reduce required wrist movement.
- Coordinated project milestones and development timelines to manage deliverables and ensure timely completion.
- Collaborated with team members to execute a KPI testing plan, evaluating the design against project objectives.

ENGINEER 1P13 Robotic End Effector | Python, Autodesk Inventor Oct. 2025 – Dec. 2025

- Designed an end effector for a Quanser's QArm to pick/drop uniquely shaped items from designated locations.
- Organized and led team meetings, assigning action items and deadlines to coordinate project timeline.
- Collaborated on mechanical and software development through CAD design and Python implementation.

Spotify Gesture Controller | Python, MediaPipe, OpenCV July 2024

- Developed a gesture based media controller for Windows using computer vision and hand gesture recognition.
- Trained a gesture classification model using MediaPipe to recognize gestures and map them to media commands.
- Designed a command system mapping distinct gestures to playback, song navigation, and volume controls.

Volunteer & Extra Curriculars

Mentor and Event Volunteer Jan. 2020 – Present
FIRST Robotics

- Supported FRC teams across Canada through mentorship and networking with established teams.
- Volunteered for FIRST Robotics events across Ontario through field setup, teardown, resets, and team queuing.
- Contributed over 500 hours to STEM outreach, robotics mentorship, and community involvement.

Technical Skills

Languages: Python, Java, R, JavaScript, HTML/CSS, C, C++
Frameworks and libraries: NumPy, pandas, OpenCV, matplotlib, Streamlit, MediaPipe, Next.js, Tailwind CSS
Tools: Onshape, Autodesk Inventor, PrusaSlicer, Git, Photoshop, Illustrator, PyCharm, Matlab, Maple, FlexPDE, Multisim